VISH SRIDHARA

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Links: Github | Website | LinkedIn | Google Scholar

SUMMARY

Bioinformatics Solutions Architect with 10+ years designing scalable NGS pipelines across **biotech, consulting, and federal organizations**, including NIH and FDA.

- Experienced in **bulk / single-cell RNA-seq, cloud workflows**, and data delivery across **AWS** and **Azure**.
- Customized nf-core/airrflow for a **3 TB, 2B+ sequence antibody dataset**, generating ESM-2-ready inputs; enhanced AWS/Snakemake analytics across **12,500+ datasets**.
- Applied AI/ML to life-sciences data using **PyTorch, TensorFlow, and MLflow**.
- Author of **20+ peer-reviewed publications**, with 1,000+ citations and an h-index of 15.

ARCHITECTURE AND TECHNICAL CAPABILITIES

- **Bioinformatics & Computing**: Nextflow, Snakemake, NGS pipelines, bulk/scRNA-seq, proteomics, HPC
- **Programming & ML**: Python (pandas, NumPy), R, SQL, Bash, PyTorch, TensorFlow, PySpark, MLflow
- **Cloud & Data**: AWS EC2/S3, Azure Databricks, Azure Data Factory, Git, CI/CD, regulated data environments
- **Data Architecture & Engineering**: Data platforms, Scalable pipelines, ETL/ELT, Workflow orchestration
- **Solution Delivery**: Rally, Requirements translation, Stakeholder collaboration, Technical documentation

EDUCATION

Postdoctoral fellow, Bioinformatics, National Institutes of Health

PHD, Electrical and Computer Engineering, Old Dominion University
Dissertation: Molecular dynamics simulations of biomolecules

MS, Electrical and Computer Engineering, Old Dominion University
Thesis: Monte-carlo analysis of GaN based semiconductors

WORK EXPERIENCE

Independent Bioinformatics Consultant, Self-employed

Jan 2026 - Present

- Streamline NGS data processing through reproducible workflows using **Python, R, and Nextflow**.
- Develop cloud **data platform** projects covering data ingestion, transformation, security, and monitoring.

Senior Bioinformatics Scientist (part-time), Argentys Informatics - *Remote*

Jul 2022 - Feb 2025 | Aug 2025 - Dec 2025

- Designed and deployed a **scalable and reproducible bioinformatics pipeline** (variant of nf-core airrflow) for antibody repertoire analysis using Observed Antibody Space Database (2B+ sequences, 3TB scale).
- Partnered with domain experts to translate ambiguous research questions into **well-scoped data science deliverables**.
- Generated **model-ready antibody repertoires** (lineage trees) for ingestion into downstream protein language model based AI workflows (ESM2 and related models).

Bioinformatics Scientist (contract), Twist Bioscience, SFO, CA

Mar 2025 - Jul 2025

- Improved an **end-to-end zero-shot inference pipeline** to predict antibody fitness and developability, defining evaluation metrics and validation strategies for robust, repeatable variant ranking.
- Worked extensively with **PyTorch** and **TensorFlow** models for training, evaluation, benchmarking, and deployment-ready inference (MLflow, AWS, Databricks).

Software Engineer (full-time), Protein Metrics LLC (Dotmatics/Siemens) - Remote
Apr 2021 - Jul 2023

- Developed command-line tools and **automated test suites** to support Byos applications.
- Supported analytics-driven software releases by executing **QA testing and creating customer-facing release notes** and product documentation in collaboration with engineering and product teams.

Senior Bioinformatics Scientist (contract), FDA, Silver Spring, MD
Jan 2019 - Apr 2021

- Built an **RNA-seq pipeline for bulk processing** of large-scale genomics datasets using fastp, HISAT2, featureCounts, and DESEQ2.
- Developed a **single-cell RNA-seq pipeline** prototype for analyzing 10X datasets using CellRanger, SAVER, Seurat/Monocle, fastMNN, and R packages.
- Characterized **immune-driven drug hypersensitivity** using single-cell RNA sequencing technologies.

Bioinformatics Software Developer (full-time), Booz Allen Hamilton, Rockville, MD
Mar 2018 - Nov 2018

- Developed PAT's **cloud-based genomics pipeline** using AWS S3, EC2, Snakemake, R, and Python to preprocess NCBI GEO data, including RNA-seq alignment, for on-demand analytics.
- Improved analytics for differential expression, gene ontology and pathway analysis for the PAT tool (**>12500 datasets** including ~20 RNA-seq projects and ~300 microarray projects).

Bioinformatics Consultant (part-time) - Remote
Apr 2016 - Apr 2018

- Protein Metrics Inc Dec 2017 - Mar 2019 - **Testing Protein Metrics suite of tools**.
- Mass General Hospital Apr 2016 - Jul 2017 - Improved primer design for a proof-of-principle **assay to capture long genomic regions**. Resulted in Nature Biomedical Engineering and other publications.

Research Eng/Sci. Associate IV (full-time), The University of Texas at Austin
Jan 2012 - Apr 2016

- Worked on **Army research office MURI grant** (multi university incl. BU, Harvard) to build mathematical models to **infer associations in multi-omics data (ML work)**. Resulted in multiple high-impact publications.
- Taught computational mass-spectrometry course in Big Data Summer School organized by CCBB (Center for Computational Biology and Bioinformatics every summer – 4 day course).

SELECTED PUBLICATIONS

Data Science, Machine Learning & Life Sciences Data - (3 of 23 Publications)

- Tosi, L., Sridhara, V., et al., Nature Biomedical Engineering. Link
- Brown, C. W., Sridhara, V., et al., BMC Genomics. Link
- Sridhara, V., et al., Proteome Science. Link

HONORS & AWARDS

NIH Postdoctoral Fellowship
Outstanding Graduate Researcher Award, Old Dominion University
BCET Scholarship, Old Dominion University

CERTIFICATIONS

AZ-104, AZ-305 (Microsoft), SAA-C03, DEA-C01 (AWS)	**In Progress**
PyTorch for Deep Learning Professional Certificate (Deeplearning.ai)	**Jan '26**
Deep Learning Specialization (Deeplearning.ai)	**Dec '24**
Microsoft Azure Databricks for Data Engineering (Coursera)	**May '24**